GETTING STARTED · BACKDROPS

Open Scratch with a fun outdoor backdrop ready to go

STEPS TO FOLLOW

1. Go to **scratch.mit.edu** and click "Create"
2. Right-click the cat sprite → Delete
3. Click "Choose a Backdrop" (bottom-right)
4. Search "Blue Sky" or "Garden" → click to add
5. Sign in to auto-save your project!

💡 TIP

Create a free Scratch account so your project saves automatically — look for "Sign in" at the top right.

🤔 DID YOU KNOW?

The Scratch stage is 480 × 360 pixels. x=0, y=0 is the very centre. Positive y = up, negative y = down.

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SPRITES · COSTUMES

Add a character sprite then customise its look in the Costumes editor

STEPS TO FOLLOW

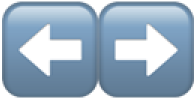
1. Click "Choose a Sprite" ⟶ pick a character
2. Click the **Costumes** tab (top-left)
3. Use the Fill tool 🪣 to recolour body parts
4. Use Brush/Shape tools to add details
5. Duplicate costume ⟶ tweak legs for walk animation
6. Click "Code" tab ⟶ set Size to 60

💡 TIP

Duplicate a costume 2–4 times with small changes to make a walking animation. Add "next costume" inside the forever loop!

🤔 DID YOU KNOW?

Every Scratch sprite image is called a "costume". Switching costumes fast creates animation — the same trick used in cartoons and flipbooks.



EVENTS · MOTION · CONTROL

Left and right arrow keys move your character — and the screen edge stops them

📜 SCRIPT REFERENCE

Walk Left & Right

SCRATCH BLOCKS

```
when 🏁 clicked
go to x: (-150) y: (-120)
forever
  if <key [left arrow] pressed?> then
    change x by (-5)
  if <key [right arrow] pressed?> then
    change x by (5)
  if on edge, bounce
```

💡 TIP

Set rotation style to "left-right" in Sprite Info so "if on edge, bounce" never flips the character upside-down!

🤔 DID YOU KNOW?

"if on edge, bounce" is a shortcut for: check x position, if past the edge reverse direction — it saves writing 4 extra blocks yourself.

LOOPS · EVENTS

Press Space to make the character rise then fall back to the ground

ADD THIS NEW SCRIPT TO YOUR CHARACTER SPRITE

```
when [space▼] key pressed
if <(y position) < (-115)> then
  repeat (10)
    change y by (12)
  end
  repeat (10)
    change y by (-12)
  end
end
```

💡 TIP

The outer "if" stops double-jumping.

Try changing 12 to 8 or 16 to make

the jump shorter or higher!

🤔 DID YOU KNOW?

This jump is linear — it rises and falls

at the same speed. Step 5 upgrades it

to feel natural, just like real games!

VARIABLES · PHYSICS

Upgrade the jump so gravity pulls the character back — just like a real platformer

SCRIPT REFERENCE

Add Gravity

REPLACE THE SPACE-KEY SCRIPT WITH THIS UPGRADE

```
when 🏁 clicked
set [y power▼] to (0)
forever
  if ‹key [space▼] pressed?› then
    if ‹(y position) < (-115)› then
      set [y power▼] to (12)
    end
  end
  change [y power▼] by (-1)
  change y by (y power)
  if ‹(y position) < (-120)› then
    set y to (-120)
    set [y power▼] to (0)
  end
end
```

💡 TIP

"change y power by (-1)" is gravity — it runs every frame so the character accelerates downward naturally.

🤔 DID YOU KNOW?

Mario, Sonic, Celeste, Hollow Knight — they all use this exact trick. You just coded real game physics!

MULTIPLE SPRITES · POSITIONING

Place an Apple sprite on the ground at a fixed position

📜 SCRIPT REFERENCE

Add the Apple Sprite

SCRATCH BLOCKS — ON THE APPLE SPRITE

```
when 🏁 clicked
go to x: (0) y: (-120)
```

💡 TIP

Always check which sprite is selected (blue outline in the sprite panel) before coding — each sprite has its own scripts!

🤔 DID YOU KNOW?

In real game engines like Unity and Godot, every character, item, and effect is a separate object with its own code — just like each Scratch sprite.

OPERATORS · RANDOM NUMBERS

Make the apple appear at a different spot each time using "pick random"

📜 SCRIPT REFERENCE

Randomize the Spawn

SCRATCH BLOCKS — ON THE APPLE SPRITE

```
when 🏁 clicked
go to x: (pick random (-180) to (180)) y: (-120)
```

💡 TIP

The y stays fixed at -120 (the ground level) — only the x is random so the apple always lands on the floor!

🤔 DID YOU KNOW?

Random number generation powers loot drops, enemy spawns, procedural maps, weather systems, and nearly every game mechanic that feels "alive".

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SENSING · FOREVER LOOP

Make the apple react with a speech bubble the moment the player touches it

SCRATCH BLOCKS — ON THE APPLE SPRITE

```
when 🏁 clicked
go to x: (pick random (-180) to (180)) y: (-120)
forever
  if <touching [your character]?> then
    say [🍎 Got it!] for (0.5) secs
```

💡 TIP

"touching?" checks pixel-by-pixel overlap 30 times per second inside the forever loop. That's called polling!

🤔 DID YOU KNOW?

Collision detection is one of the most important algorithms in games. Unity calls them "colliders", Godot calls them "collision shapes" — same idea.

VARIABLES · STATE

Create a Score variable that increases by 1 each time an apple is collected

📜 SCRIPT REFERENCE

SCRATCH BLOCKS — ON THE APPLE SPRITE

```
when 🏁 clicked
set [Score] to (0)
go to x: (pick random (-180) to (180)) y: (-120)
forever
  if <touching [your character]?> then
    change [Score] by (1)
```

💡 TIP

Tick the checkbox next to "Score" in the Variables palette to show/hide the score display on stage. Right-click → "Large readout" for big numbers!

🤔 DID YOU KNOW?

A variable is the most fundamental idea in all programming. Python, JavaScript, Swift, C++ — every language uses them to remember and update information.

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GAME LOOP · RESPAWN PATTERN

After collecting, teleport the apple to a new random spot so the game never ends

📜 SCRIPT REFERENCE

Respawn the Apple

SCRATCH BLOCKS — ON THE APPLE SPRITE

```
when 🏁 clicked
set [Score] to (0)
go to x: (pick random (-180) to (180)) y: (-120)
forever
  if <touching [your character]?> then
    change [Score] by (1)
    go to x: (pick random (-180) to (180)) y: (-120)
```

💡 TIP

Game complete! Try adding more apples, a countdown timer, or different point values to increase the challenge.

🤔 DID YOU KNOW?

"Game loop" is this pattern: check input → update state → render → repeat. Every game in history — from Pong to Fortnite — runs on this same cycle.